

## Advanced (Habanero)

- Q1.** Which of the following is a condition for a function to have an inverse?
- (A) The function must be one-to-one  
(B) The function must be a quadratic  
(C) The function must be a polynomial  
(D) The function must be periodic  
(E) The function must be exponential
- Q2.** What is the inverse of the function  $f(x) = 2x + 1$ ?
- (A)  $f^{-1}(x) = \frac{x-1}{2}$   
(B)  $f^{-1}(x) = \frac{x+1}{2}$   
(C)  $f^{-1}(x) = 2x - 1$   
(D)  $f^{-1}(x) = \frac{x-2}{1}$   
(E)  $f^{-1}(x) = x^2 + 1$
- Q3.** To verify if  $f(x) = x^2$  has an inverse, which of the following must be true?
- (A)  $f(x) = x$   
(B)  $f(x) = -x$   
(C)  $f(x)$  must be one-to-one  
(D)  $f(x)$  must be a linear function  
(E)  $f(x)$  must be a constant function
- Q4.** If  $f(x) = 3x - 2$  and  $g(x) = \frac{x+2}{3}$ , which of the following is true?
- (A)  $f(g(x)) = x$  and  $g(f(x)) = x$   
(B)  $f(g(x)) = x$  but  $g(f(x)) \neq x$   
(C)  $g(f(x)) = x$  but  $f(g(x)) \neq x$   
(D)  $f(g(x)) \neq x$  and  $g(f(x)) \neq x$   
(E)  $f(g(x)) = g(f(x))$
- Q5.** The function  $f(x) = e^x$  has an inverse because it is
- (A) a polynomial function  
(B) a rational function  
(C) a one-to-one function  
(D) a periodic function  
(E) a quadratic function
- Q6.** The inverse of  $f(x) = \log_2 x$  is
- (A)  $f^{-1}(x) = 2^x$   
(B)  $f^{-1}(x) = \log_2 x$   
(C)  $f^{-1}(x) = x^2$   
(D)  $f^{-1}(x) = \frac{1}{x}$   
(E)  $f^{-1}(x) = x$
- Q7.** If  $f(x) = x^3 + 1$ , which of the following is the inverse function?
- (A)  $f^{-1}(x) = \sqrt[3]{x-1}$   
(B)  $f^{-1}(x) = \sqrt[3]{x+1}$   
(C)  $f^{-1}(x) = x^3 - 1$   
(D)  $f^{-1}(x) = \frac{x-1}{3}$   
(E)  $f^{-1}(x) = x - 1$
- Q8.** What is the domain of the inverse function  $f^{-1}(x) = \sqrt{x}$ ?
- (A)  $[0, \infty)$   
(B)  $(-\infty, 0]$   
(C)  $(-\infty, \infty)$   
(D)  $[0, 1]$   
(E)  $(-1, 1)$

### Open Ended Questions (FRQ)

- Q9.** Find the inverse of the function  $f(x) = \frac{1}{x+1}$  and verify using composition. Show all your work and explain your reasoning.
- Q10.** A city's population  $P$  (in thousands) can be modeled by the function  $P(t) = 2t + 5$ , where  $t$  is the time in years. Find the inverse function and interpret its meaning in the context of the problem. Show all your work and explain your reasoning.

### Chef's Tips

- Remind students to show all work and explain their reasoning for FRQs
- Encourage students to use proper mathematical notation and formatting
- Emphasize the importance of verifying inverses using composition